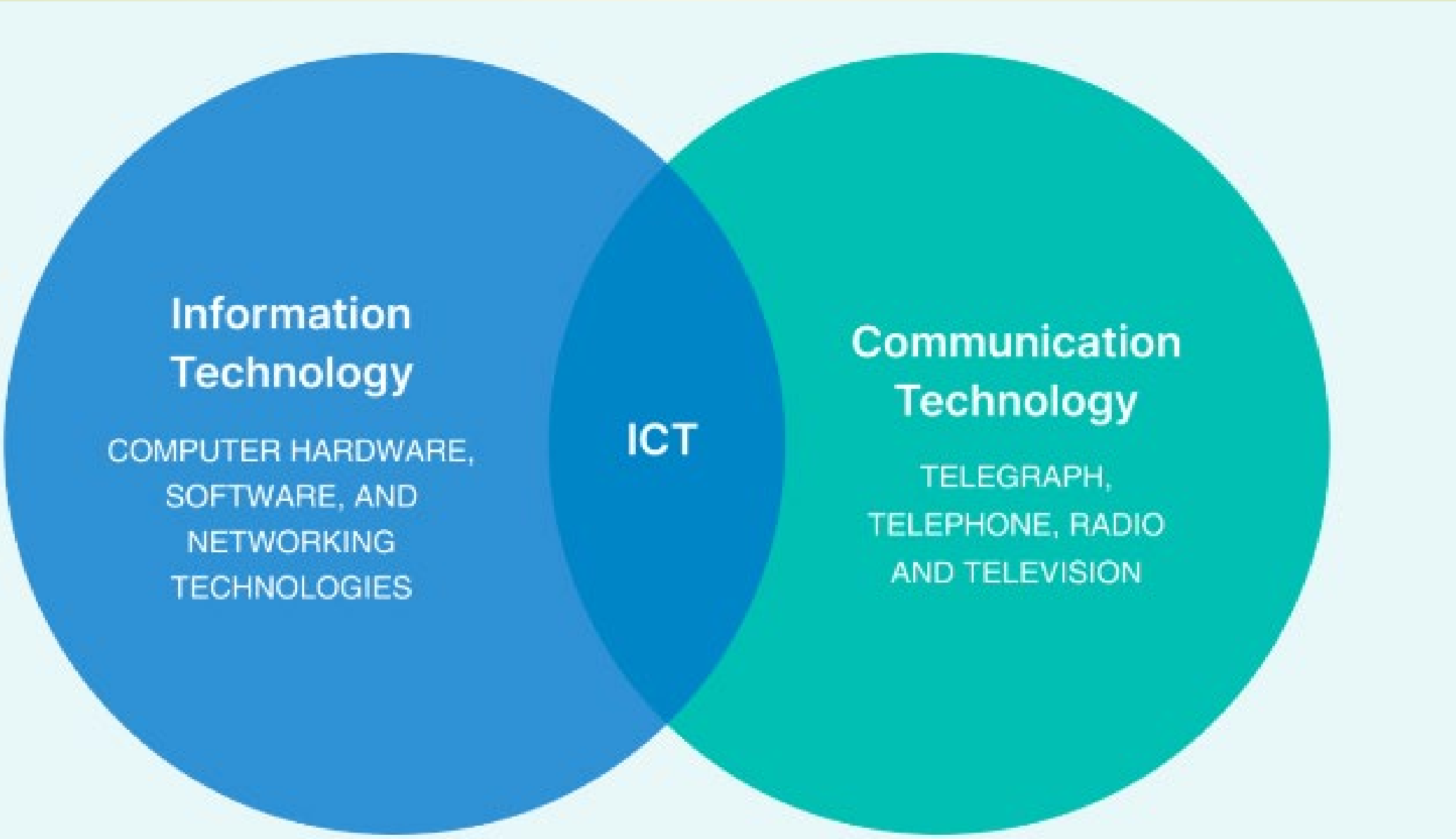
ICT Infrastructure and Systems


Information Technology
COMPUTER HARDWARE, SOFTWARE, AND NETWORKING TECHNOLOGIES
ICT
Communication Technology
TELEGRAPH, TELEPHONE, RADIO AND TELEVISION

ICT Infrastructure and Systems Includes:

- Hardware and software that supports the way information is created, distributed, acquired and stored.
- Infrastructure and electronics that enable communication between hardware devices.
- Protocols and interfaces that enable seamless communication and data exchanges between different hardware and software components.

Continues…

- Tools for protecting sensitive information and ensuring the integrity of an ICT system.
- Standards for protecting data in transit, during processing and at rest.
- Governance policies for how information should be accessed, secured, processed, transmitted and stored.
- Workers who have the skills required to design, develop, maintain and support ICT systems

ICT Infrastructure

Information Technology	Telecommunications	ICT Services
Hardware	Wired Communication	Internet / Cloud Computing
Software	Wireless Communication	Social Networking
Data Storage	VoIP	AI and ML
Data Governance	Unified Communication	IoT

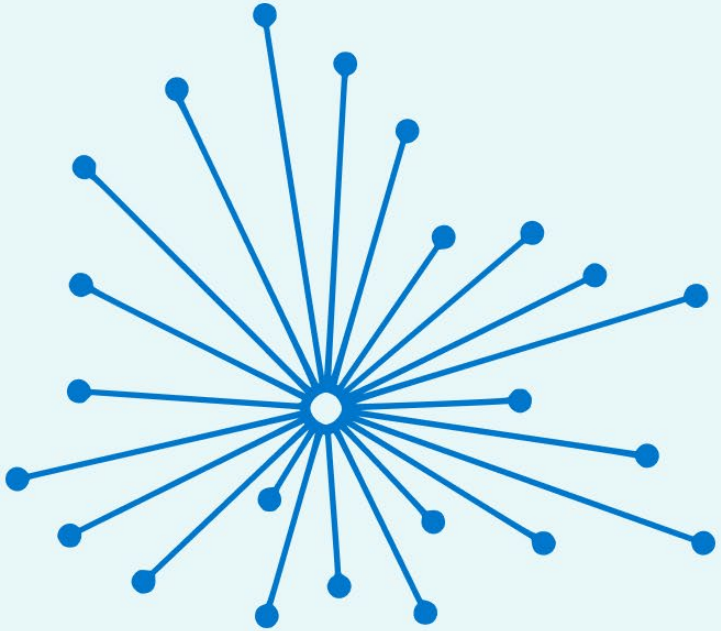
Centralized ICT

- When ICT architectures are centralized, the hardware, software and network infrastructure components that support communication are located in a single location and users can only access these resources from that location.
- This approach was cost-efficient during the early days of computing when the cost of wired communication infrastructure was high and mainframes had limited processing power.

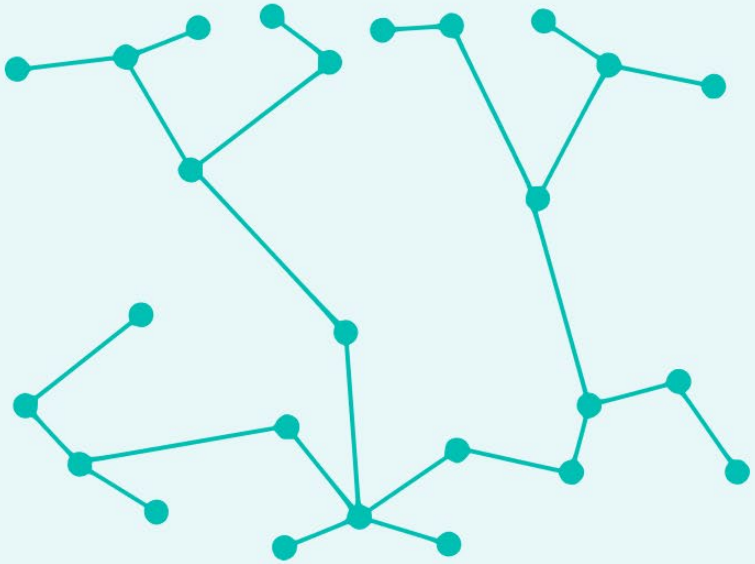
Decentralized ICT

- Broadband internet made it possible to decentralize ICT architectures. Once data could be transmitted faster and over greater distances, it became possible to store and access data remotely and run the same applications virtually on multiple remote servers. This led to increased adoption of unified communication systems (UCSs), cloud-based services and new types of collaboration tools.
- The increased bandwidth broadband provides has also inspired new ways to use peer-to-peer (P2P) (skype) network architectures. In a P2P network, each node (or user) can act as both a server and a client and share information with other nodes without the need for a central server.

Centralized ICT



Decentralized ICT



Emerging Trends in ICT

Artificial Intelligence (AI)

- Artificial Intelligence continues to rank at the top of emerging tech. Chabot's and Apple's Siri are common examples of AI in action, but we will see more complex versions being used in the near future.

Internet of Things (IoT)

- Internet of Things (IoT) uses connected devices, such as wearables and sensors, to provide more optimal results for businesses.

Continues…

Machine Learning (ML)

- Machine learning has helped technology advance significantly in recent years, giving us the ability to draw insights from massive amounts of data.

5G Networks

- 5G networks have been key for increasing speeds, increasing capacity and reducing latency. The next wave of networks will involve 5G networks that stand alone.

Continues…

Edge Computing

- According to Deloitte, U.S. households have an average of 22 connected devices. Edge computing may well be the answer to managing big data.

Blockchain

- Blockchain is helping to keep data more secure and creating a reliable history of transactions. The applications extend far beyond that of cryptocurrency.

A-1: Task-1

- **Read and search more about these emerging trends. Prepare a table of all these trends, their applications, and impact on society. Also highlight which one is your favorite, and why?**